

STUDENT NAME: _____

TEACHER: _____



HURLSTONE
AGRICULTURAL
HIGH
SCHOOL

2021

HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK 4

Mathematics Extension 1

Examiners

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**General
Instructions**

- Preparation time – 10 minutes
- Working time – 2 hours
- Scanning and uploading time – 1 hour
- Write using black pen
- Calculators approved by NESA may be used
- A Reference sheet is provided at the back of this paper
- Multiple Choice Answer sheet is provided at the back of this paper
- For questions in Section II, complete on your own paper and show relevant mathematical reasoning and/or calculations

**Total marks:
70****Section I – 10 marks**

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 60 marks

- Attempt Questions 11 – 14
- Allow about 1 hour and 45 minutes for this section

SECTION I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section.

Use the multiple-choice answer sheet for Questions 1 – 10

Question 1

What is the domain of the function $y = 5 \cos^{-1}(4x)$?

- A. $-\frac{5}{4} \leq x \leq \frac{5}{4}$
- B. $-\frac{1}{5} \leq x \leq \frac{1}{5}$
- C. $-\frac{1}{4} \leq x \leq \frac{1}{4}$
- D. $-1 \leq x \leq 1$

Question 2

Which of the following gives the coefficient of x^4 in the expansion of $(x - \sqrt{2})^6$?

- A. -30
- B. 30
- C. -60
- D. 60

Question 3

Let $P(x) = x^2 + bx + c$ where b and c are constants. The zeroes of $P(x)$ are α and $\alpha + 1$. What are the correct expressions for b and c in terms of α ?

- A. $b = -(2\alpha + 1)$ and $c = \alpha^2 + \alpha$
- B. $b = 2\alpha + 1$ and $c = \alpha^2 + \alpha$
- C. $b = \alpha^2 + \alpha$ and $c = -(2\alpha + 1)$
- D. $b = \alpha^2 + \alpha$ and $c = 2\alpha + 1$

Question 4

The rate of change of the population (P) of a small regional centre in NSW can be approximated by the formula $\frac{dP}{dt} = -\frac{1}{10}(P - 10000)$.

Which one of the equations below is a solution to this differential equation?

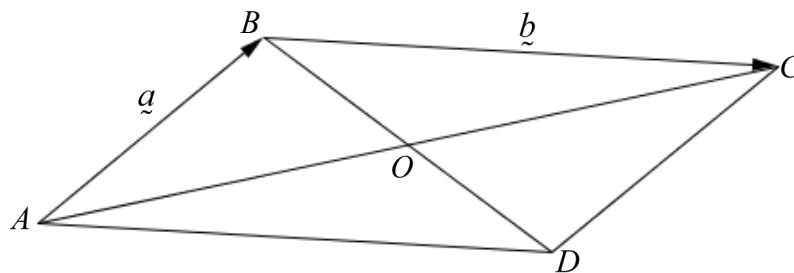
- A. $P = 10000e^{-0.1t}$
- B. $P = -10000e^{-0.1t}$
- C. $P = 10000 - 2000e^{-0.1t}$
- D. $P = 10000 + 2000e^{0.1t}$

Question 5

Which of the following is an expression for $y = \sin x + \sqrt{3} \cos x$ in the form $R \sin(x + \alpha)$?

- A. $2 \sin\left(x + \frac{\pi}{4}\right)$
- B. $2 \sin\left(x + \frac{\pi}{3}\right)$
- C. $2 \sin\left(x - \frac{\pi}{4}\right)$
- D. $2 \sin\left(x - \frac{\pi}{3}\right)$

Question 6



In the parallelogram $ABCD$ shown, the point of intersection of the diagonals is O .

$$\overrightarrow{AB} = \tilde{a} \text{ and } \overrightarrow{BC} = \tilde{b}$$

What is the vector $\overrightarrow{OC}$ equal to?

- A. $\frac{1}{2}(\tilde{a} - \tilde{b})$
- B. $\frac{1}{2}(\tilde{a} + \tilde{b})$
- C. $\frac{1}{2}\tilde{a} - \tilde{b}$
- D. $\frac{1}{2}\tilde{b} - \tilde{a}$

Question 7

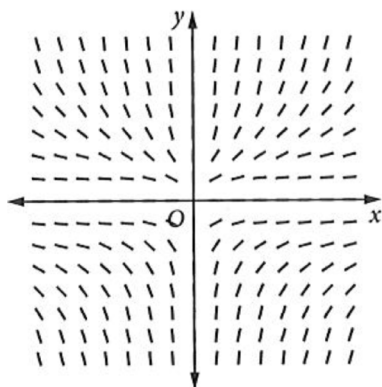
If $y = \sin^{-1} \frac{a}{x}$, which of the following is an expression for $\frac{dy}{dx}$?

- A. $\frac{-a}{x^2 \sqrt{x^2 - a^2}}$
- B. $\frac{x}{\sqrt{x^2 - a^2}}$
- C. $\frac{-x}{\sqrt{x^2 - a^2}}$
- D. $\frac{-a}{x \sqrt{x^2 - a^2}}$

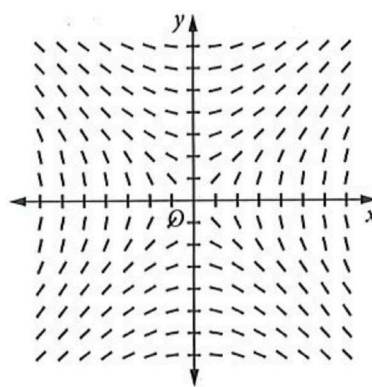
Question 8

Which one of the slope fields below could be used to represent $xy' - y = 0$?

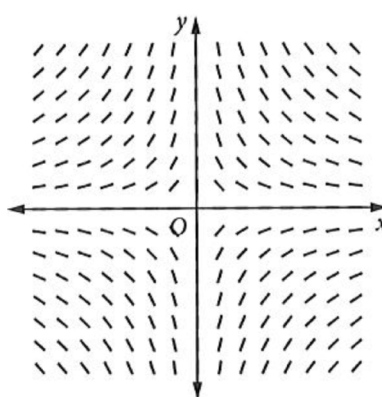
A.



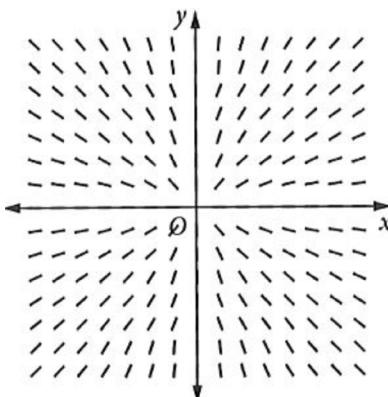
B.



C.



D.



Question 9

Which of the following is a true statement?

A. $\sin 3x \sin 4y = \frac{1}{2} [\cos(3x - 4y) - \cos(3x + 4y)]$

B. $\sin 3x \sin 4y = \frac{1}{2} [\cos(3x + 4y) - \sin(3x - 4y)]$

C. $\sin 3x \sin 4y = \frac{1}{2} [\cos(3x + 4y) - \cos(3x - 4y)]$

D. $\sin 3x \sin 4y = \frac{1}{2} [\cos(3x - 4y) - \sin(3x + 4y)]$

Question 10

What is the unit vector in the direction of $\underline{a} = -2\underline{i} + 5\underline{j}$?

A. $\frac{1}{7}(-2\underline{i} + 5\underline{j})$

B. $\frac{1}{29}(-2\underline{i} + 5\underline{j})$

C. $\frac{1}{\sqrt{29}}(-2\underline{i} + 5\underline{j})$

D. $\frac{1}{\sqrt{21}}(-2\underline{i} + 5\underline{j})$

End of Section I

SECTION II

60 marks

Attempt Questions 11 – 14

Allow about 1 hour and 45 minutes for this section.

Answer each question on your own paper.

For questions in **Section II**, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a new Writing Booklet

- (a) Four politicians and four activists are seated alternately at a round table. **1**
(i.e. alternating so that no two politicians or two activists sit next to each other)
In how many ways can this be done?
- (b) The letters of the word CONSIDER are rearranged at random. What is the probability that **1**
an arrangement begins with 'SI' ?
- (c) A selection from the integers from 1 to 8 is made. How many numbers need to be selected **1**
from this group in order to be certain that the selection contains a pair of numbers that add
up to 10?

Question 11 continued on Page 8

- (d) Two boats on a lake start sailing at the same time.

Boat A moves on a course given by: $x = \frac{t}{2}$; $y = t + 1$.

Boat B moves on a course given by: $x = t - 2$: $y = 9 - 2t$.

where t measures the time elapsed in hours from when the boats started.

- (i) Find the Cartesian equation for the course of each boat and show that the courses intersect at $(1, 3)$. 2

- (ii) Do the boats collide? Justify your answer supported by calculations. 1

- (e) Find the vector projection of $\underline{a}$ onto $\underline{b}$ if $\underline{a} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\underline{b} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$. 2

- (f) Find $\underline{d}$ perpendicular to $\underline{c} = 4\underline{i} - 3\underline{j}$ with a magnitude of 10. 3

- (g) $\triangle ABC$ is a triangle with $\overrightarrow{AB} = \underline{c}$ and $\overrightarrow{AC} = \underline{b}$.

D and E are the midpoints of $\overrightarrow{AB}$ and $\overrightarrow{AC}$ respectively.

F is a point on $\overrightarrow{BC}$ such that $\overrightarrow{FC} = 2 \times \overrightarrow{BF}$

- (i) Express the vectors $\overrightarrow{BC}$ and $\overrightarrow{DE}$ in terms of $\underline{b}$ and $\underline{c}$. 1

- (ii) What geometric property of a triangle is demonstrated by the answers to (i)? 1

- (iii) Express $\overrightarrow{AF}$ in terms of $\underline{b}$ and $\underline{c}$. 2

End of Question 11

Question 12 (15 marks) Use a new Writing Booklet

- (a) Consider the function $f(x) = x^2 - 4x + 6$.
- (i) Explain why the domain of $f(x)$ must be restricted if $f(x)$ is to have an inverse function. 1
- (ii) Given that the domain of $f(x)$ is restricted to $x \leq 2$, find an expression for $f^{-1}(x)$. 2
- (b) Differentiate $e^x \tan^{-1}(x)$. 2
- (c) Find $\int_0^{\pi} \frac{4}{\sqrt{16-x^2}} dx$. 2
- (d) Find $\int 2 \cos^2(x) dx$. 2
- (e) Using the substitution $u = x^2 - 2$, or otherwise, find the exact value of $\int_2^3 \frac{x}{\sqrt{x^2-2}} dx$. 3
- (f) Prove by mathematical induction that $4^n + 14$ is divisible by 6 for all positive integers n where $n \geq 1$. 3

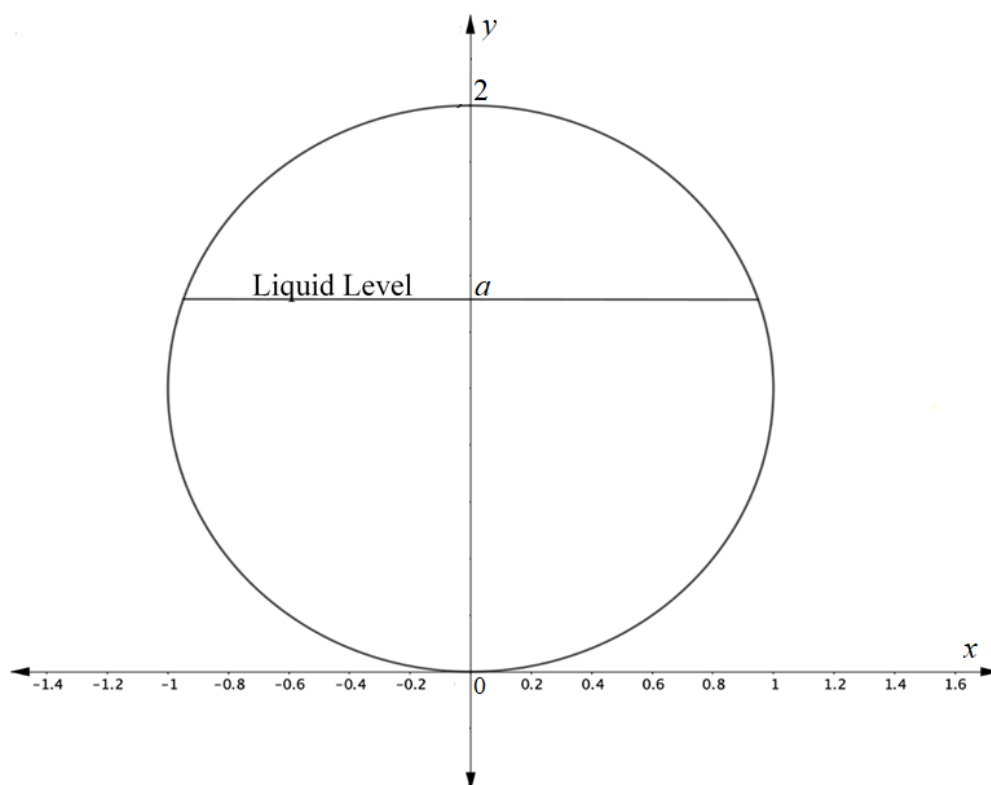
End of Question 12

Question 13 (15 marks) Use a new Writing Booklet

- (a) A spherical tank used to transport an industrial liquid has a radius of 1 metre.

The volume of liquid in the tank for a depth of a metres can be found by calculating the volume of a suitable solid of revolution using integration methods.

The diagram below shows the cross section of the tank through its centre drawn on a cartesian plane.



- (i) Show that the volume of liquid, of depth a , in the tank is given by:

3

$$V = \frac{\pi a^2}{3}(3 - a)$$

where a is measured in metres.

- (ii) The tank in (i) above is being filled at a constant rate of $\frac{\pi}{10}$ m³/min.

(α) Find the rate of change of volume (V) with respect to depth (a).

1

(β) At what rate is the depth of the liquid increasing when the depth is 0.4 m?

2

Question 13 continued on Page 11

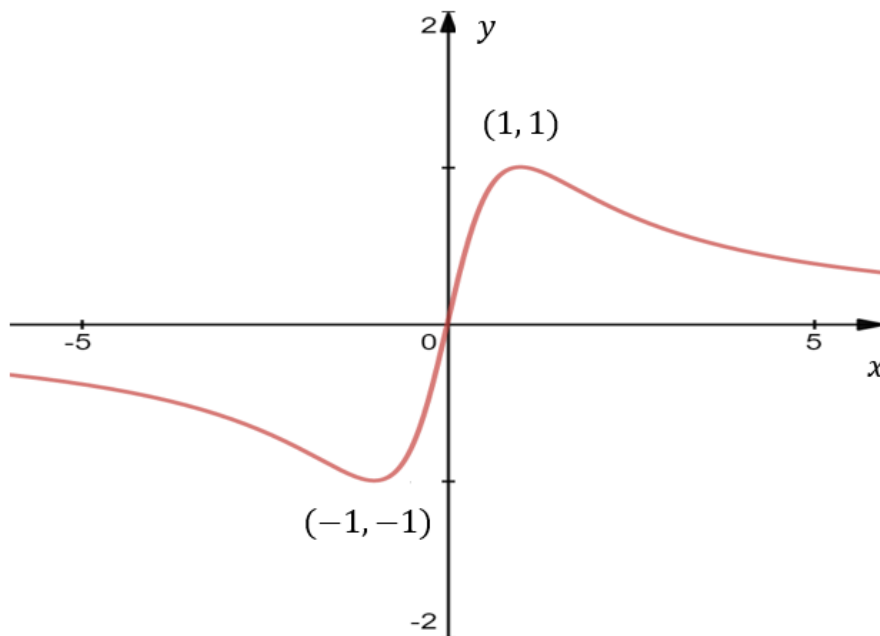
- (b) The region bounded by the curve $y = e^{-x}$, the coordinate axes and the line $x = b$, $b > 0$, is rotated about the x -axis.
- (i) Find the volume of the solid of revolution generated in terms of b . 2
- (ii) Find the limit of the volume of this solid as $b \rightarrow \infty$. 1
- (c) Solve the differential equation $\frac{dy}{dx} = \cos^2 y$ given that $y = \frac{\pi}{4}$ when $x = 0$. 2
- (d) The formula for the height of a tree, H , after t years can be found by solving the differential equation $\frac{dH}{dt} = 0.03H(45 - H)$, given the tree is 0.3 metres high when it is planted.
- (i) Show that the expression $\frac{1}{H(45 - H)} = \frac{1}{H} + \frac{1}{45 - H}$. 1
- (ii) Solve the differential equation to find the formula for the height of the tree. 3

End of Question 13

Question 14 (15 marks) Use a new Writing Booklet

- (a) The graph of the function $y = \frac{2x}{(x^2 + 1)}$ is shown below.

1



Sketch the graph of $y = \frac{2|x|}{(x^2 + 1)}$. Label all important features.

- (b) By dividing the polynomial $f(x) = 2x^4 - 10x^3 + 12x^2 + 2x - 3$ by

2

$g(x) = x^2 - 3x + 1$ write $f(x)$ in the form shown below:

$f(x) = g(x)q(x) + r(x)$ where $q(x)$ and $r(x)$ are polynomials

and $r(x)$ has degree less than 2.

Question 14 continued on Page 13

- (c) Given that $(\sin \alpha - \cos \alpha)^2 = 1 - \sin 2\alpha$, What is the exact value of $\sin 75^\circ - \cos 75^\circ$? 2

Leave your answer with a rational denominator.

- (d) (i) If $\sin(x + \theta) = A \sin(x - \theta)$, prove that $(A - 1) \tan x = (A + 1) \tan \theta$. 3

- (ii) Hence solve $\sin(x + 40^\circ) = 3 \sin(x - 40^\circ)$, for $0^\circ \leq x \leq 360^\circ$. 2

- (e) (i) Prove that $\tan(A + B + C) = \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A}$. 2

- (ii) Given A, B and C are acute angles of a triangle and $\frac{\tan A}{5} = \frac{\tan B}{6} = \frac{\tan C}{7} = k$.

3

Hence or otherwise show that $k = \sqrt{\frac{3}{35}}$ is the only solution to $\tan(A + B + C) = 0$.

End of Paper

Year 12 2021 HSC Assessment Task 4 Marking Guidelines

Answers to Multiple Choice questions

Question 1:

$$y = 5\cos^{-1}(4x)$$

$$-1 \leq 4x \leq 1$$

$$-\frac{1}{4} \leq x \leq \frac{1}{4}$$

∴ Answer C

Question 2:

The term that contains x^4 in the expansion of $(x - \sqrt{2})^6$ is $6C_4x^4(-\sqrt{2})^2$

$$\text{Coefficient of } x^4 = 6C_4(-\sqrt{2})^2 = 15 \times 2 = 30$$

∴ Answer B

Question 3:

Since the zeroes of $P(x)$ are α and $\alpha + 1$

$$\text{Sum of roots: } \alpha + \alpha + 1 = -\frac{b}{1} \quad \text{Product of roots}$$

$$2\alpha + 1 = -b \quad \alpha \times (\alpha + 1) = \frac{c}{1}$$

$$\therefore b = -(2\alpha + 1) \quad \therefore c = \alpha^2 + \alpha$$

∴ Answer A

Question 4:

Upon differentiating

$$P = 10000 - 2000e^{-0.1t}$$

$$\frac{dP}{dt} = 200e^{-0.1t}$$

$$= \frac{1}{10}(2000e^{-0.1t})$$

$$\text{but, } 2000e^{-0.1t} = 10000 - P$$

$$\begin{aligned} \therefore \frac{dP}{dt} &= \frac{1}{10}(10000 - P) \\ &= -\frac{1}{10}(P - 10000), \text{ the required differential equation} \end{aligned}$$

∴ Answer C

Question 5:

$$R = \sqrt{1^2 + (\sqrt{3})^2} = 2$$

$$\tan \alpha = \frac{\sqrt{3}}{1}, \quad \alpha = \tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$$

$$\therefore \sin x + \sqrt{3} \cos x = 2 \sin\left(x + \frac{\pi}{3}\right)$$

$\therefore$ Answer B

Question 6:

Diagonals of a parallelogram bisect each other. $\therefore \overline{OC} = \frac{1}{2} \overline{AC}$

$\therefore$ Answer B

Question 7:

$$\begin{aligned} y &= \sin^{-1} \frac{a}{x} \\ \frac{dy}{dx} &= \frac{1}{\sqrt{1 - \left(\frac{a}{x}\right)^2}} \times -\frac{a}{x^2} \\ &= \frac{-a}{x\sqrt{x^2 - a^2}} \end{aligned}$$

$\therefore$ Answer D

Question 8:

The differential equation $xy' - y = 0$ can be rearranged to give $y' = \frac{y}{x}$.

When $x > 0$ and $y > 0$, $y' > 0$. This rules out option C as gradients are negative in the first quadrant.

Also, when $x < 0$ and $y < 0$, $y' > 0$. This rules out A as gradients are negative in the third quadrant.

Lastly, when $x = 0$, gradients will be undefined. This leaves D as the correct answer as tangents are vertical when $x = 0$ and B has horizontal tangents when $x = 0$.

$\therefore$ Answer D

Question 9:

From reference sheet using $\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$

$\sin 3x \sin 4y = \frac{1}{2} [\cos(3x - 4y) - \cos(3x + 4y)]$ is a true statement

$\therefore$ **Answer A**

Question 10:

All 4 responses feature the same vector $\mathbf{a}$ in brackets.

To get unit vector, it's divided by its length. Use Pythagoras to do so.

$\therefore$ **Answer C**

Year 12 Mathematics Extension 1 Assessment Task 4 2021

Question No. 11		Solutions and Marking Guidelines
Outcomes Addressed in this Question		
ME 11-1: uses algebraic and graphical concepts in the modelling and solving of problems involving functions		
ME 11-5: uses concepts of permutations and combinations to solve problems involving counting or ordering		
ME 12-2: Applies concepts and techniques involving vectors to solve problems		
Outcome	Solutions	Marking Guidelines
ME 11-5	(a) $3! \times 4! = 144$	(a) Award 1: Correct answer.
	(b) $\frac{6!}{8!}$ or $\frac{1}{8} \times \frac{1}{7} = \frac{1}{56}$	(b) Award 1: Correct answer.
	(c) 5 pigeonholes: [1], [5], [2,8], [3,7], [4,6] Therefore 6 values need to be selected to be sure of a pair that add to 10.	(c) Award 1: Correct answer.
ME 11-1	(d) (i) Boat A: $y = 2x + 1$ Boat B: $y = 5 - 2x$ Substitute $x = 1$ to RHS to show that LHS = 3 = y Boat A: $2(1) + 1 = 3$ Boat B: $5 - 2(1) = 3$ Therefore the point of intersection is (1, 3). (ii) Boat A is at (1, 3) when $t=2$, i.e. after 2 hours. Boat B is at (1, 3) when $t=3$, i.e. after 3 hours. Therefore they do not collide.	(d) (i) Award 2: Correct solution Award 1 Both equations correct Award 1 Justification of point of intersection. (ii) Award 1: Correct answer with justification.
ME 12-2	(e) $\text{Projection} = \frac{1 \times 2 + 2 \times 2}{(2 \times 2)^2} \times \begin{pmatrix} 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 1\frac{1}{2} \\ 1\frac{1}{2} \end{pmatrix}$ (f) Let $\underline{d} = x\underline{i} + y\underline{j} \rightarrow$ dot product: $4x - 3y = 0$ $ \underline{d} = 10 \rightarrow x^2 + y^2 = 100$ $\rightarrow x = 6, -6 \quad y = 8, -8$ $\therefore \underline{d} = \pm(6\underline{i} + 8\underline{j})$ Alternately, you could find the 2 vectors in the form $\underline{d} = k \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ where $ \underline{d} ^2 = 100$ (g) (i) $\overrightarrow{BC} = \underline{b} - \underline{c}$ $\overrightarrow{DE} = \frac{1}{2}(\underline{b} - \underline{c})$ (ii) The interval joining the midpoints of 2 sides of a triangle is parallel to and half the length of the third side.	(e) Award 2 Correct solution. Award 1 At least one correct dot product, or equivalent. (f) Award 3: Correct solution with working. Award 2: Error in working, or failure to find both solutions. Award 1: Some significant progress. (g) (i) Award 1 both vectors correct. (ii) Award 1: Both properties mentioned

	(iii) $\overrightarrow{AF} = \overrightarrow{AB} + \overrightarrow{BF} = \underline{c} + \frac{1}{3}\overrightarrow{BC}$ $= \frac{1}{3}\underline{b} + \frac{2}{3}\underline{c}$	(iii) Award 2: Correct solution simplified to one component of <i>b</i> and <i>c</i>. Award 1: Some significant progress.
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Year 12	Mathematics Extension 1 2021	TASK 4
Question No. 12	Solutions and Marking Guidelines	
Outcomes Addressed in this Question		
ME12-1 –	applies techniques involving proof or calculus to model and solve problems	
ME11-1 –	uses algebraic and graphical concepts in the modelling and solving of problems involving functions and their inverses	
Part / Outcome	Solutions	Marking Guidelines
(a)(i) ME11-1	$f(x) = x^2 - 4x + 6$ is a parabola, and <u>fails the horizontal line test</u> . So, the inverse of $f(x)$ over its entire domain is not a function.	1 mark – Correct solution (explains using the horizontal line test, or equivalent merit)
(a)(ii) ME11-1	Complete the square to express $f(x)$ in vertex form: $f(x) = x^2 - 4x + 6 \quad (x \leq 2)$ $= (x - 2)^2 + 2$ Swap x and y, and make y the subject: $x = (y - 2)^2 + 2 \quad (y \leq 2)$ $x - 2 = (y - 2)^2$ $y - 2 = -\sqrt{x - 2} \quad \left(\text{discard } +\sqrt{x - 2} \text{ as } y \leq 2\right)$ $y = -\sqrt{x - 2} + 2$ $f^{-1}(x) = -\sqrt{x - 2} + 2$	2 marks – Correct solution 1 mark – Substantially correct (swaps x and y, or equivalent merit)
(b) ME12-1	$\frac{d}{dx}(e^x \tan^{-1} x) = u'v + v'u$ $= e^x \tan^{-1} x + \frac{e^x}{1 + x^2}$	2 marks – Correct solution 1 mark – Substantially correct (uses product rule)
(c) ME12-1	$\int_0^\pi \frac{4}{\sqrt{16 - x^2}} dx = 4 \left[\sin^{-1} \left(\frac{x}{4} \right) \right]_0^\pi$ $= 4 \left[\sin^{-1} \left(\frac{\pi}{4} \right) - \sin^{-1}(0) \right]$ $= 4 \sin^{-1} \left(\frac{\pi}{4} \right) \quad (\approx 3.61)$	2 marks – Correct solution 1 mark – significant progress towards correct solution (eg finds correct integral) *** see note below about this question

***Please note:

(c) Many candidates demonstrated a lack of awareness that

$\frac{\pi}{4}$ is not an angle in this question

$$\sin^{-1}\left(\frac{\pi}{4}\right) \neq \frac{1}{\sqrt{2}} \dots \text{detail is important}$$

Similarly:

(e) Some candidates did not change the x limits at the substitution stage. When the substitution is made, the limits should also change to be in terms of u . Even if later changing back to be in terms of x .

Detail is important.

Year 12 Mathematics Extension 1 Task 4 2021		
Question No. 13 Solutions and Marking Guidelines		
Outcomes Addressed in this Question		
MA12-7 applies the concepts and techniques of indefinite and definite integrals in the solution of problems		
Outcome	Solutions	Marking Guidelines
ME12-7	(a) (i) Equation of circular cross section: $x^2 + (y-1)^2 = 1$ $x^2 = 1 - (y-1)^2$ $= 1 - y^2 + 2y - 1$ $= 2y - y^2$ Volume generated by rotating the curve about the y-axis between 0 and a (ie. the depth of the liquid in the tank) $V = \pi \int_a^b x^2 dy$ $= \pi \int_0^a 2y - y^2 dy$ $= \pi \left[y^2 - \frac{y^3}{3} \right]_0^a$ $= \pi \left(a^2 - \frac{a^3}{3} \right)$ $= \pi \left(\frac{3a^2 - a^3}{3} \right)$ $= \frac{\pi a^2}{3} (3 - a) \text{ as required}$	3 marks Correct solution showing equation of circular cross section, primitive function and substitution leading to required expression. 2 marks Substantial progress towards correct solution. 1 mark Some progress towards correct solution showing knowledge of the process of finding the volume of a solid of revolution.
ME12-7	(ii) (α) $V = \frac{\pi a^2}{3} (3 - a)$ $= \pi a^2 - \frac{\pi a^3}{3}$ $\frac{dV}{da} = 2\pi a - \pi a^2$ $= \pi a (2 - a)$	1 mark Correct answer.
ME12-7	(β) Given $\frac{dV}{dt} = \frac{\pi}{10}$ Rate of change of depth with respect to time $\frac{da}{dt} = \frac{da}{dV} \cdot \frac{dV}{dt}$ $= \frac{1}{\pi a (2 - a)} \cdot \frac{\pi}{10}$ $= \frac{1}{10a(2 - a)}$ When depth of liquid $a = 0.4$ $\frac{da}{dt} = \frac{1}{10 \times 0.4 (2 - 0.4)}$ $= \frac{1}{4 \times 1.6}$ $= \frac{5}{32} \text{ m / min}$	2 marks Correct solution making use of the chain rule. 1 mark Substantial progress towards correct solution.

ME12-7	(b) (i) $y = e^{-x}$ $y^2 = e^{-2x}$ $V = \pi \int_a^b y^2 dx$ $= \pi \int_0^b e^{-2x} dx$ $= \pi \left[-\frac{1}{2} e^{-2x} \right]_0^b$ $= \pi \left[\left(-\frac{1}{2} e^{-2b} \right) - \left(-\frac{1}{2} e^0 \right) \right]$ $= \pi \left(-\frac{1}{2} e^{-2b} + \frac{1}{2} \right)$ $= \frac{\pi}{2} (1 - e^{-2b}) \text{ units}^3$	2 marks Correct solution. 1 mark Substantial progress towards correct solution.
ME12-7	(ii) $\lim_{b \rightarrow \infty} \frac{\pi}{2} (1 - e^{-2b}) = \frac{\pi}{2} (1 - 0) \quad \text{since } \lim_{x \rightarrow \infty} (e^{-x}) = 0$ $= \frac{\pi}{2} \text{ units}^3$	1 mark Correct evaluation of the limit.
ME12-7	(c) $\frac{dy}{dx} = \cos^2 y$ $\frac{dx}{dy} = \frac{1}{\cos^2 y}$ $= \sec^2 y$ $x = \tan y + c$ When $y = \frac{\pi}{4}$, $x = 0$ $\therefore 0 = \tan \frac{\pi}{4} + c$ $c = -\tan \frac{\pi}{4}$ $= -1$ $\therefore x = \tan y - 1$ $x + 1 = \tan y$ $y = \tan^{-1}(x + 1)$	2 marks Correct solution. 1 mark Substantial progress towards correct solution.
ME12-7	(d) (i)	1 mark Correct solution

ME12-7

$$\begin{aligned}\text{RHS} &= \frac{1}{H} + \frac{1}{45-H} \\ &= \frac{(45-H)}{H(45-H)} + \frac{H}{H(45-H)} \\ &= \frac{45-H+H}{H(45-H)} \\ &= \frac{45}{H(45-H)}\end{aligned}$$

OR

$$\frac{1}{H(45-H)} = \frac{1}{45} \left(\frac{1}{H} + \frac{1}{45-H} \right)$$

(ii)

$$\begin{aligned}\frac{dH}{dt} &= 0.03H(45-H) \\ \frac{dt}{dH} &= \frac{1}{0.03H(45-H)} \\ &= \frac{1}{0.03 \times 45} \left(\frac{1}{H} + \frac{1}{45-H} \right) \\ \int \frac{dt}{dH} dH &= \frac{1}{1.35} \int \frac{1}{H} + \frac{1}{45-H} dH \\ t &= \frac{1}{1.35} [\ln|H| - \ln|45-H|] + c \\ t - c &= \frac{1}{1.35} \ln \left| \frac{H}{45-H} \right| \\ 1.35(t - c) &= \ln \left| \frac{H}{45-H} \right| \\ e^{1.35(t-c)} &= \frac{H}{45-H}, 0 \leq H \leq 45 \\ e^{1.35t} \times e^{-1.35c} &= \frac{H}{45-H} \\ Ae^{1.35t} &= \frac{H}{45-H}, \text{ where } A = e^{-1.35c} \\ \text{When } t = 0, H = 0.3 \\ A &= \frac{0.3}{45-0.3} \\ &= \frac{1}{149} \\ \frac{1}{149} e^{1.35t} &= \frac{H}{45-H} \\ e^{1.35t} &= \frac{149H}{(45-H)} \\ (45-H)e^{1.35t} &= 149H \\ 45e^{1.35t} &= 149H + e^{1.35t}H \\ &= H(149 + e^{1.35t}) \\ H &= \frac{45e^{1.35t}}{149 + e^{1.35t}} \\ &= \frac{45}{149e^{-1.35t} + 1} \\ &= \frac{45}{1 + 149e^{-1.35t}}\end{aligned}$$

3 marks

Correct solution showing all required steps.

2 marks

Substantial progress towards correct solution, including showing correct use of equality shown to be true in part (i).

1 mark

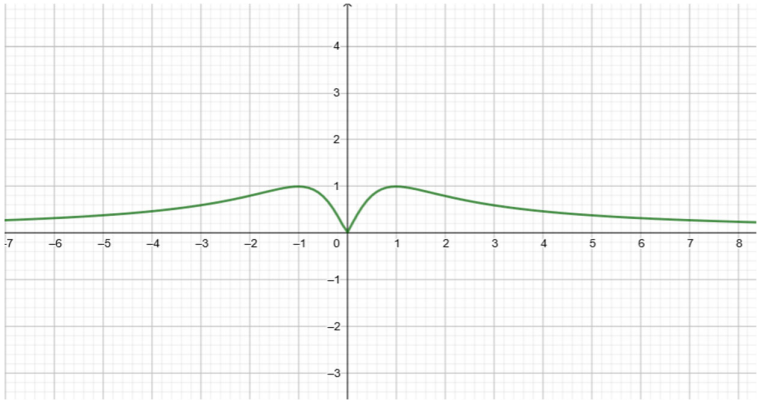
Some progress towards correct solution.

Outcomes Addressed in this Question

ME11-1 : uses algebraic and graphical concepts in the modelling and solving of problems involving functions and their inverses

ME11-2: manipulates algebraic expressions and graphical functions to solve problems

ME12-3: applies advanced concepts and techniques in simplifying expressions involving compound angles and solving trigonometric equations

Outcome	Solutions	Marking Guidelines
ME11-1	(a) 	Award 1 mark for the correct graph.
ME11-2	(b) $ \begin{array}{r} 2x^2 - 4x - 2 \\ x^2 - 3x + 1 \overline{) 2x^4 - 10x^3 + 12x^2 + 2x - 3} \\ \underline{2x^4 - 6x^3 + 2x^2} \\ 0 - 4x^3 + 10x^2 + 2x - 3 \\ \underline{-4x^3 + 12x^2 - 4x} \\ 0 - 2x^2 + 6x - 3 \\ \underline{-2x^2 + 6x - 2} \\ -1 \end{array} $ $\therefore 2x^4 - 10x^3 + 12x^2 + 2x - 3 = [x^2 - 3x + 1][2x^2 - 4x - 2] - 1$	Award 2 marks for the correct solution. Award 1 mark for substantial progress towards the solution
ME12-3	(c) Using $(\sin \alpha - \cos \alpha)^2 = 1 - \sin 2\alpha$ (given) $ \begin{aligned} (\sin 75^\circ - \cos 75^\circ)^2 &= 1 - \sin(2 \times 75^\circ) \\ &= 1 - \sin 150^\circ \\ &= 1 - \sin(180^\circ - 30^\circ) = 1 - \sin 30^\circ = 1 - \frac{1}{2} = \frac{1}{2} \end{aligned} $ $ \sin 75^\circ - \cos 75^\circ = \pm \sqrt{\frac{1}{2}} = \pm \frac{1}{\sqrt{2}} $ $\therefore \sin x \geq \cos x$ for $45^\circ \leq x \leq 90^\circ$ $ \therefore \sin 75^\circ - \cos 75^\circ = \sqrt{\frac{1}{2}} = \frac{1}{\sqrt{2}} $	Award 2 marks for the correct solution. Award 1 mark for substantial progress towards the solution

ME12-3	(d)(i) Using $\sin(x + \theta) = A \sin(x - \theta)$ $\sin x \cos \theta + \cos x \sin \theta = A[\sin x \cos \theta - \cos x \sin \theta]$ $\sin x \cos \theta + \cos x \sin \theta = A \sin x \cos \theta - A \cos x \sin \theta$ $A \cos x \sin \theta + \cos x \sin \theta = A \sin x \cos \theta - \sin x \cos \theta$ $(A + 1) \cos x \sin \theta = (A - 1) \sin x \cos \theta$ divide both sides by $\cos x \cos \theta$ $\frac{(A + 1) \cos x \sin \theta}{\cos x \cos \theta} = \frac{(A - 1) \sin x \cos \theta}{\cos x \cos \theta}$ $\frac{(A + 1) \sin \theta}{\cos \theta} = \frac{(A - 1) \sin x}{\cos x}$ $(A + 1) \tan \theta = (A - 1) \tan x$ $\therefore (A - 1) \tan x = (A + 1) \tan \theta$ (d)(ii) $\sin(x + 40^\circ) = 3 \sin(x - 40^\circ)$ using part (i) $(3 - 1) \tan x = (3 + 1) \tan 40^\circ$ $2 \tan x = 4 \tan 40^\circ$ $\tan x = 2 \tan 40^\circ$ $x = \tan^{-1}(2 \tan 40^\circ), 180^\circ + \tan^{-1}(2 \tan 40^\circ)$ $x = 59.21^\circ, 239.21^\circ$ (correct to two decimal places) (e) (i)	Award 3 marks for the correct solution. Award 2 mark for substantial progress towards the correct solution. Award 1 mark for some progress towards the correct solution. Award 2 marks for the correct solution. Award 1 mark for substantial progress towards the solution
ME12-3	$\tan(A + B + C) = \tan\left[A + (B + C)\right] = \frac{\tan A + \tan(B + C)}{1 - \tan A \tan(B + C)}$ $= \frac{\tan A + \frac{\tan B + \tan C}{1 - \tan B \tan C}}{1 - \tan A \left[\frac{\tan B + \tan C}{1 - \tan B \tan C} \right]}$ $= \frac{\frac{\tan A(1 - \tan B \tan C) + \tan B + \tan C}{1 - \tan B \tan C}}{1 - \tan B \tan C - \tan A(\tan B + \tan C)}$ $= \frac{\tan A - \tan A \tan B \tan C + \tan B + \tan C}{1 - \tan B \tan C - \tan A \tan B - \tan A \tan C}$ $= \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A}$	Award 2 marks for the correct solution. Award 1 mark for substantial progress towards the solution

ME12-3	(e) (ii) In $\triangle ABC$, $A + B + C = 180^\circ$ $\therefore \tan(A + B + C) = \tan 180^\circ = 0$ using part (i) $\frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A} = 0$ $\tan A + \tan B + \tan C - \tan A \tan B \tan C = 0$ $\tan A + \tan B + \tan C = \tan A \tan B \tan C \quad \text{-----} \boxed{A}$ Given $\frac{\tan A}{5} = \frac{\tan B}{6} = \frac{\tan C}{7} = k$ $\therefore \tan A = 5k, \tan B = 6k$ and $\tan C = 7k$ using $\boxed{A}$ $5k + 6k + 7k = 5k \times 6k \times 7k$ $18k = 210k^3$ $210k^3 - 18k = 0$ $6k(35k^2 - 3) = 0$ $6k = 0, k^2 = \frac{3}{35}$ $\therefore k = 0, k = \pm \sqrt{\frac{3}{35}}$ The only valid value is $k = +\sqrt{\frac{3}{35}}$ since $\tan A > 0, \tan B > 0$ and $\tan C > 0$ and A, B and C are acute.	Award 3 marks for the correct solution. Award 2 mark for substantial progress towards the correct solution. Award 1 mark for some progress towards the correct solution.
ME12-3		